VISUALIZING US NATURAL DIASTER DECLARATION – TRENDS AND PATTERNS

Week 1 Documentation/ Screenshots

Data Cleaning in Power BI

Data Cleaning in Power BI

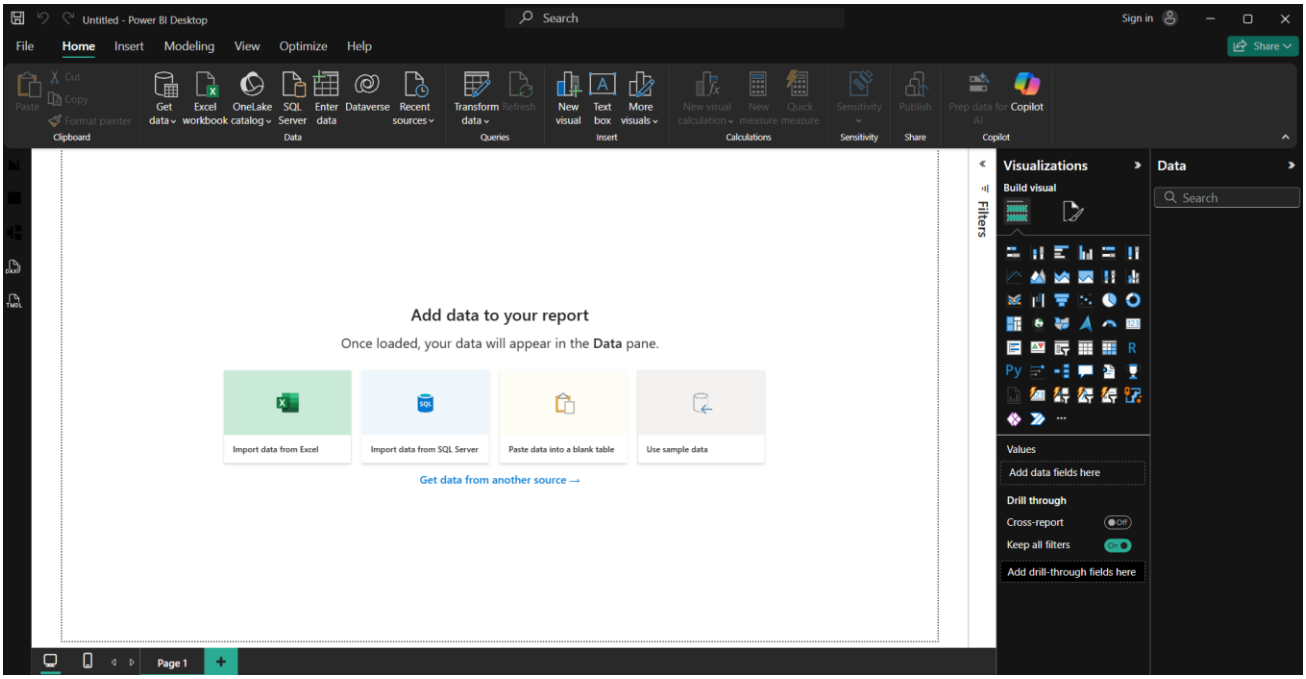


Fig 1.1 PowerBI Canvas

Load Data:

Get Data -> Text/CSV -> Choose File -> Transform Data

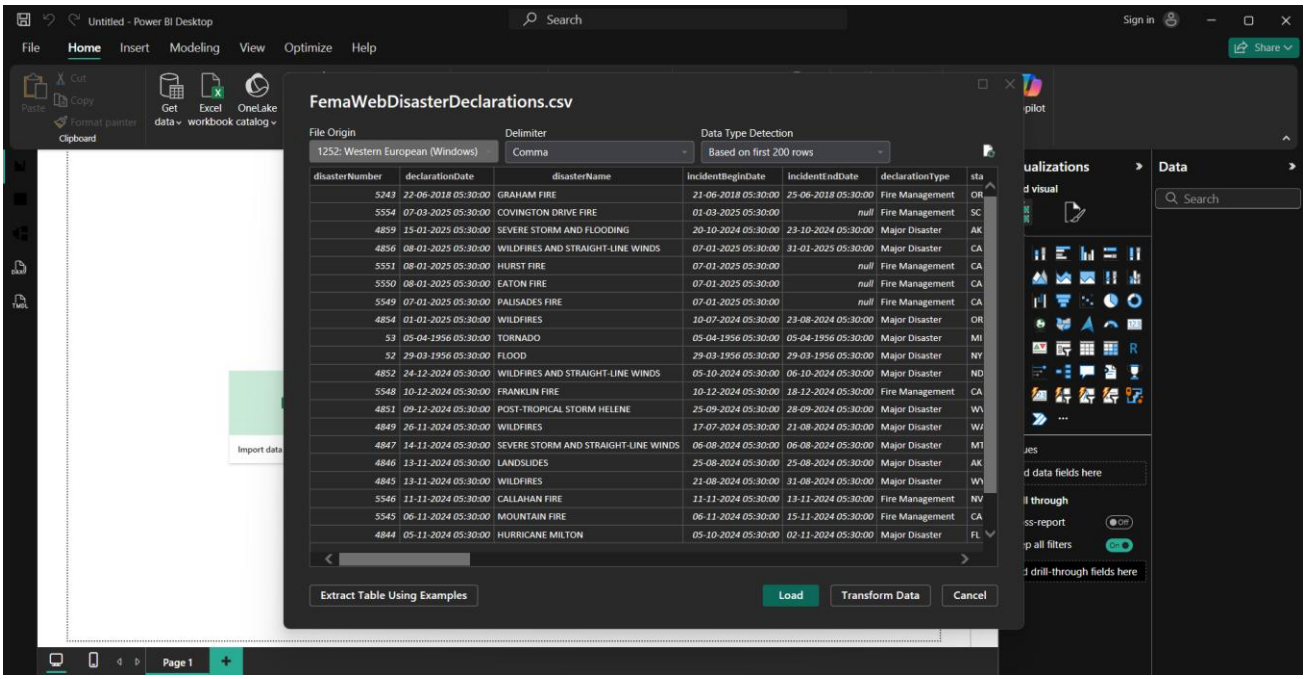


Fig 1.2 Transform Data

Remove Duplicate Rows:

Ctrl+A -> Remove Rows -> Remove Duplicates

Remove Unnecessary Columns:

Click StateCode column, Ctrl+alt+(other columns to remove) -> Remove Columns (in toolbar ribbon) -> Remove Columns

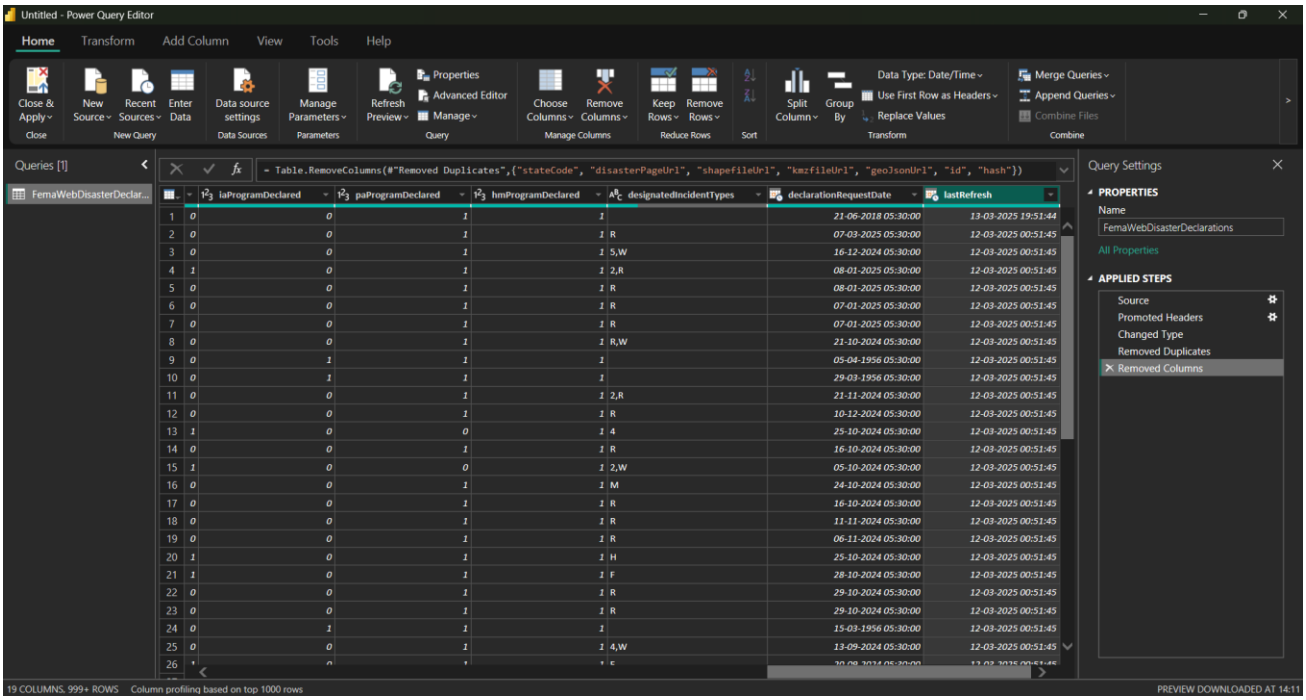


Fig 1.3 Removed unnecessary columns

Check Datatype:

To convert disaster programs declared to Boolean: Click the number icon -> true/false

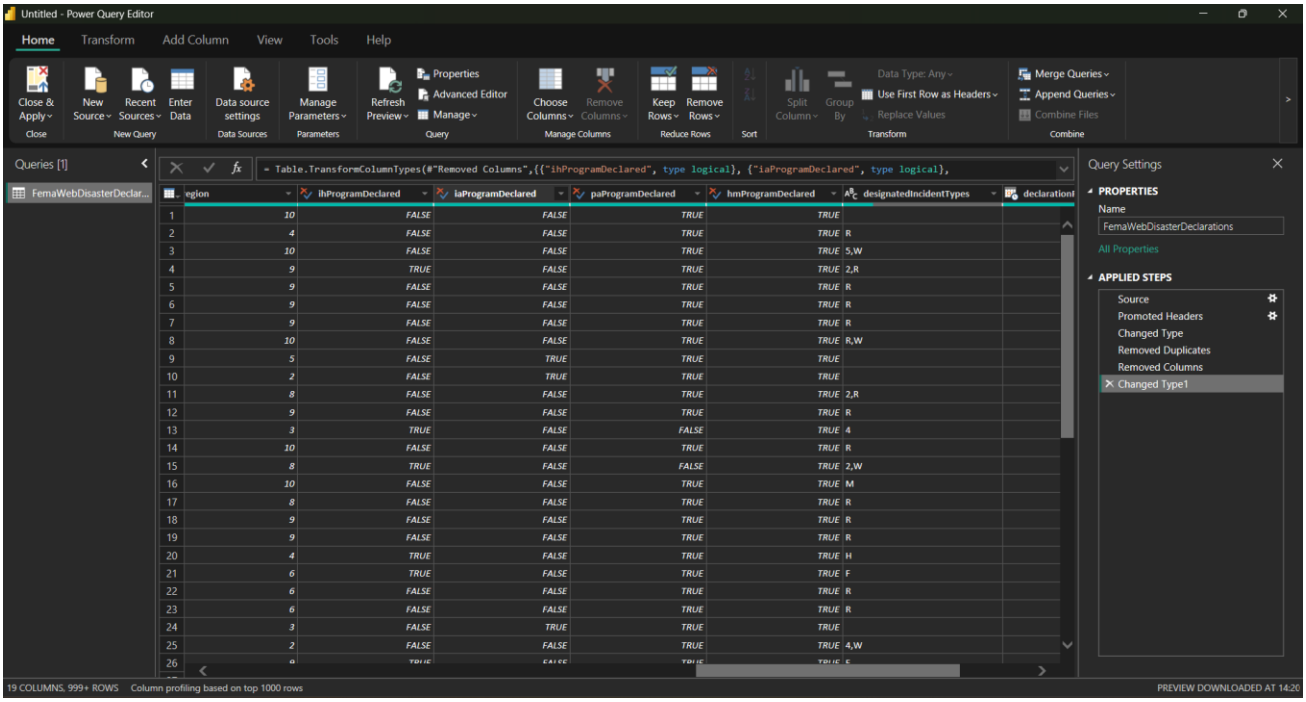


Fig 1.4 Changed Datatype

Replace blank with Unknown: (in designatedIncidentType)

Select column -> replace values -> , Unknown

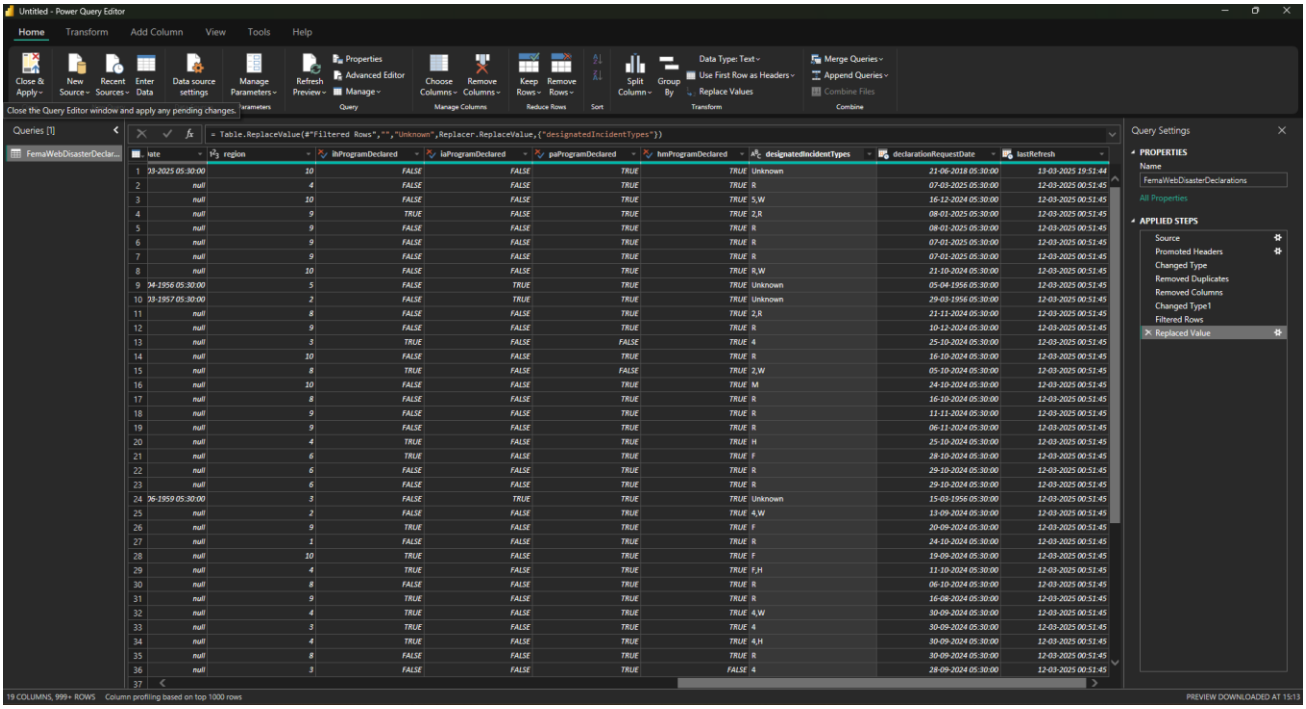


Fig 1.5 Replaced blank values

Adding Useful Columns:

1. Status

Creating Conditional Columns:

Select column -> Add Column -> Conditional Column

Using closeOutDate:

Create a column named “Status” which is used to depict whether the disaster is official declared as closed or not.

IF (closeOutDate EQUALS null) THEN Status = “Open”

ELSE Status = “Closed”

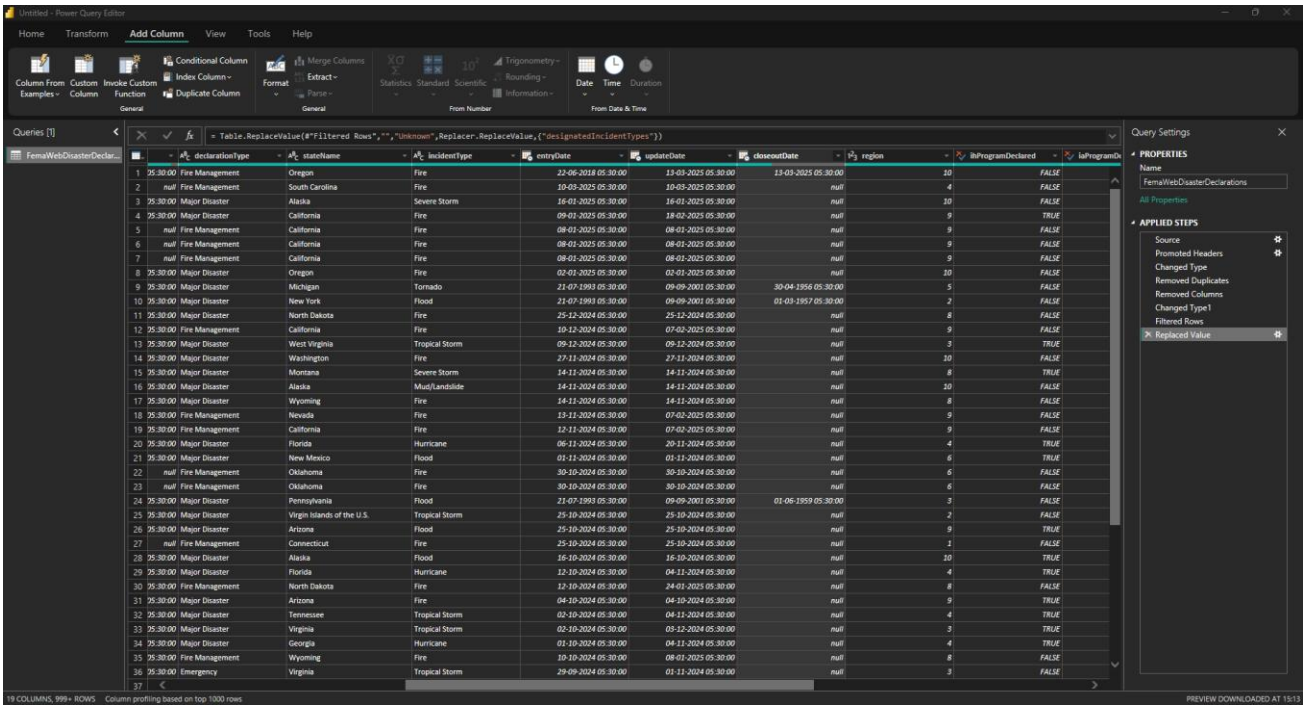


Fig 1.6 Before creating “Status” column

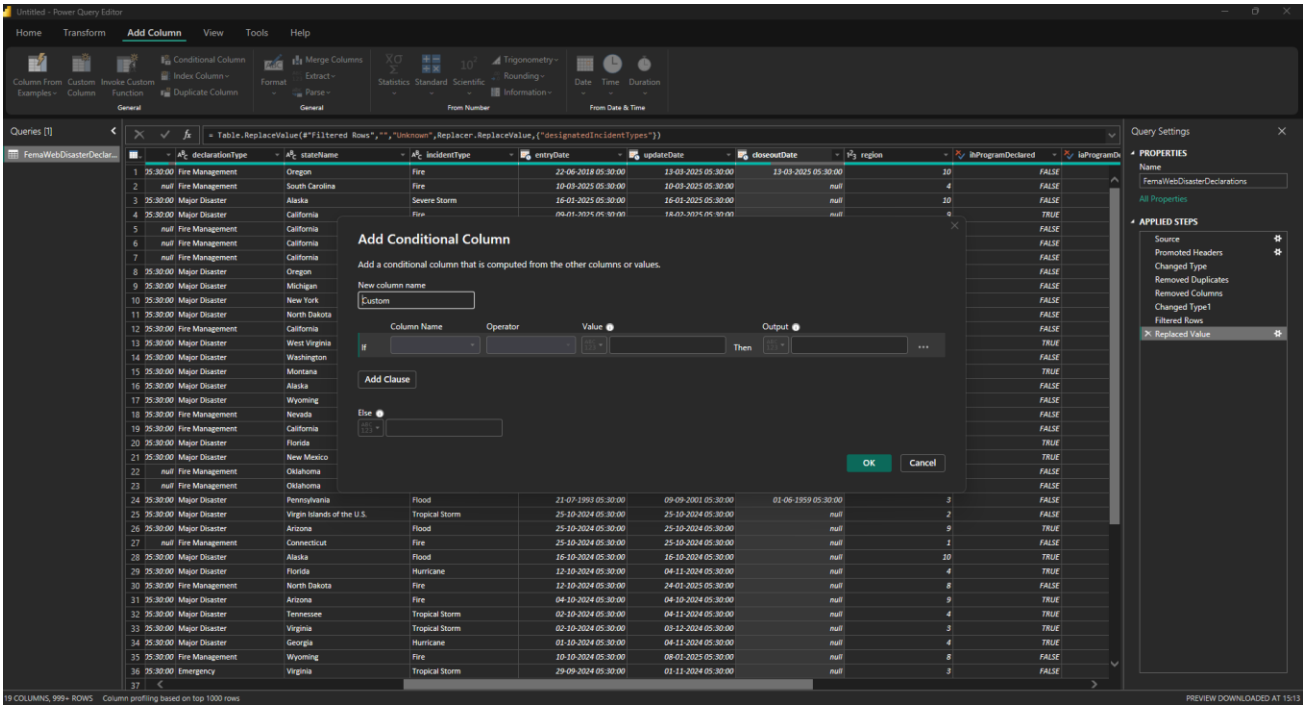


Fig 1.7 “Add Conditional Column” pop-up

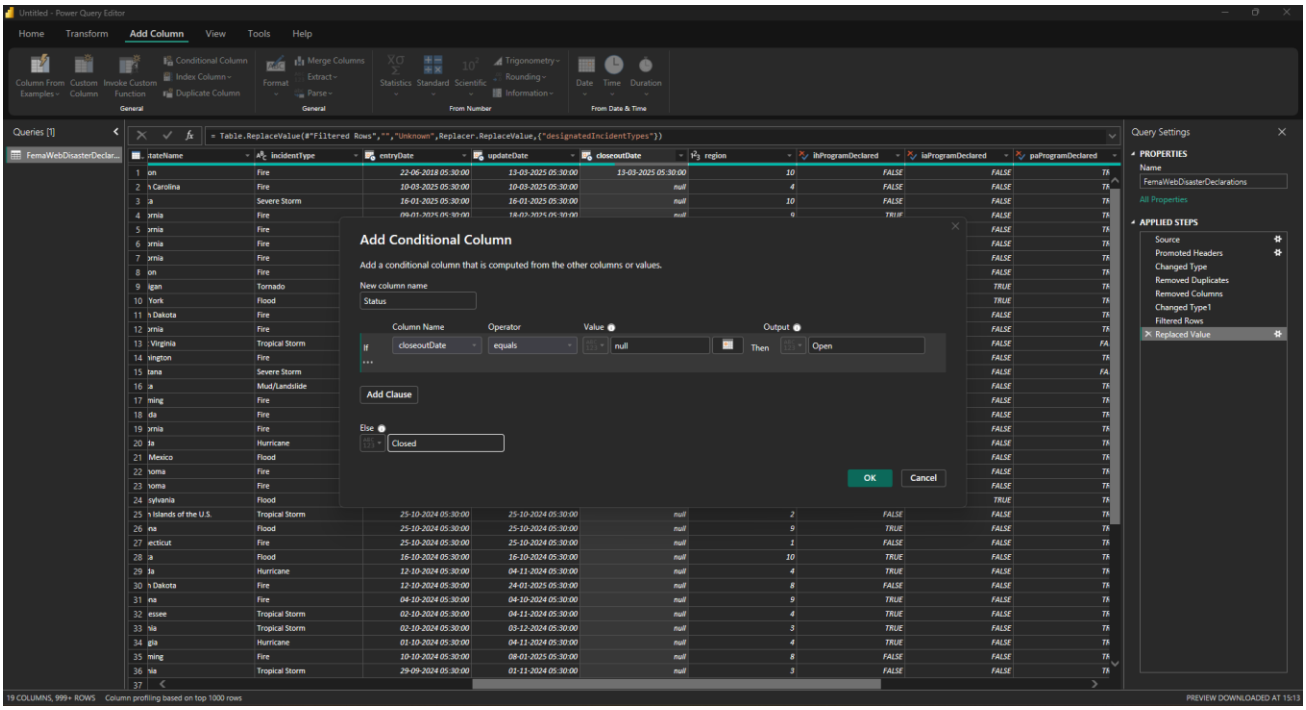


Fig 1.8 “Add Conditional Column” pop-up

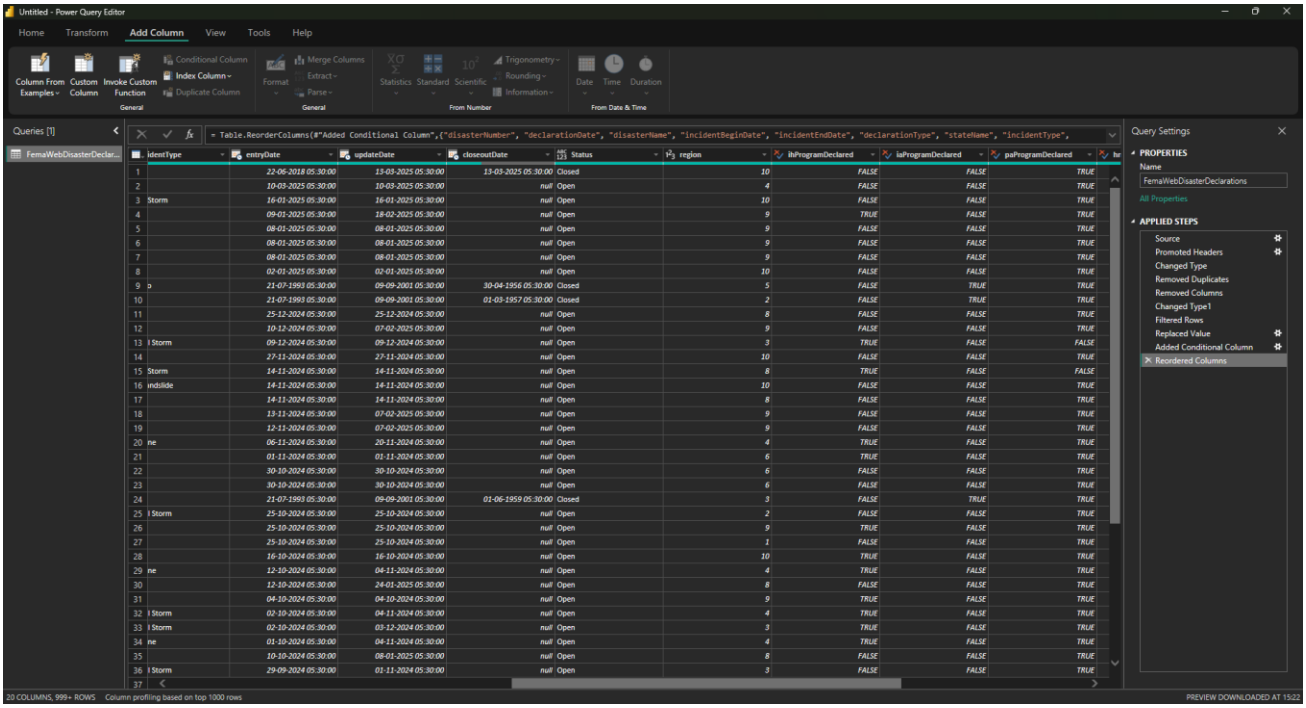


Fig 1.9 Created a conditional column “Status”

2. Incident Duration

= Duration.Days([incidentEndDate] - [incidentBeginDate])

In header, Add Column -> Custom Column

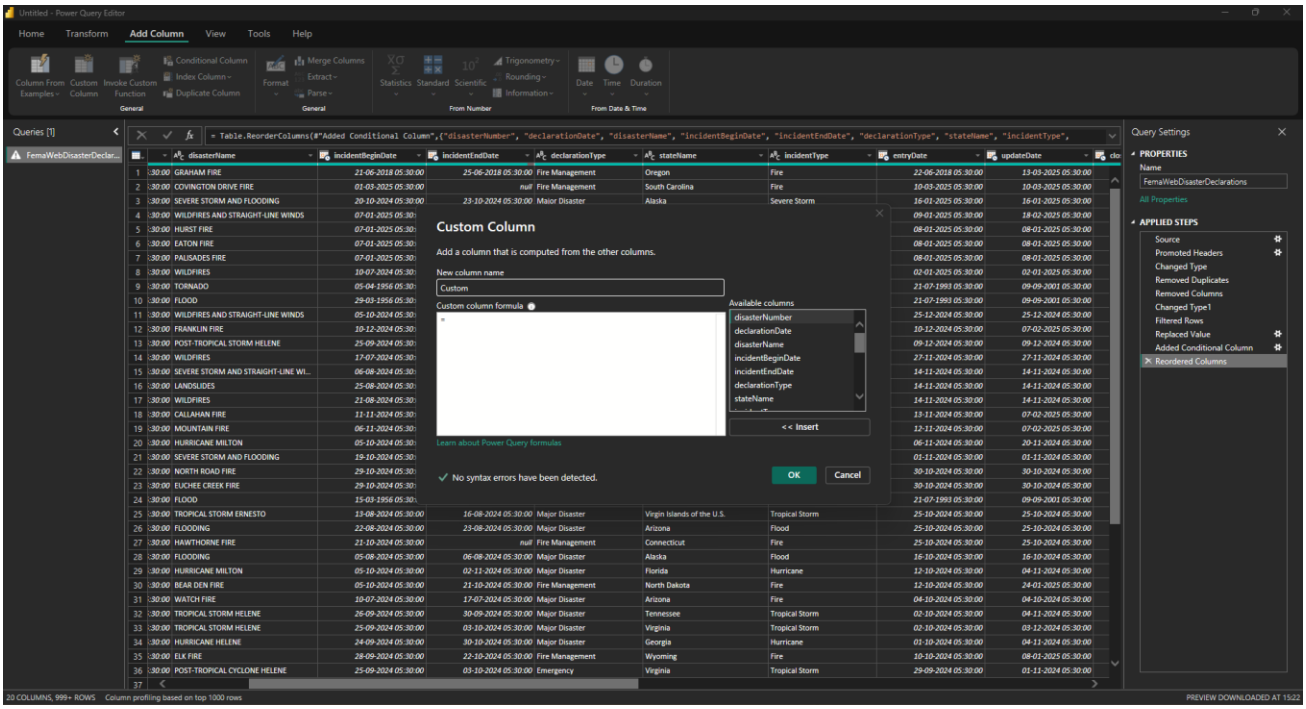


Fig 1.10 Custom Column pop-up

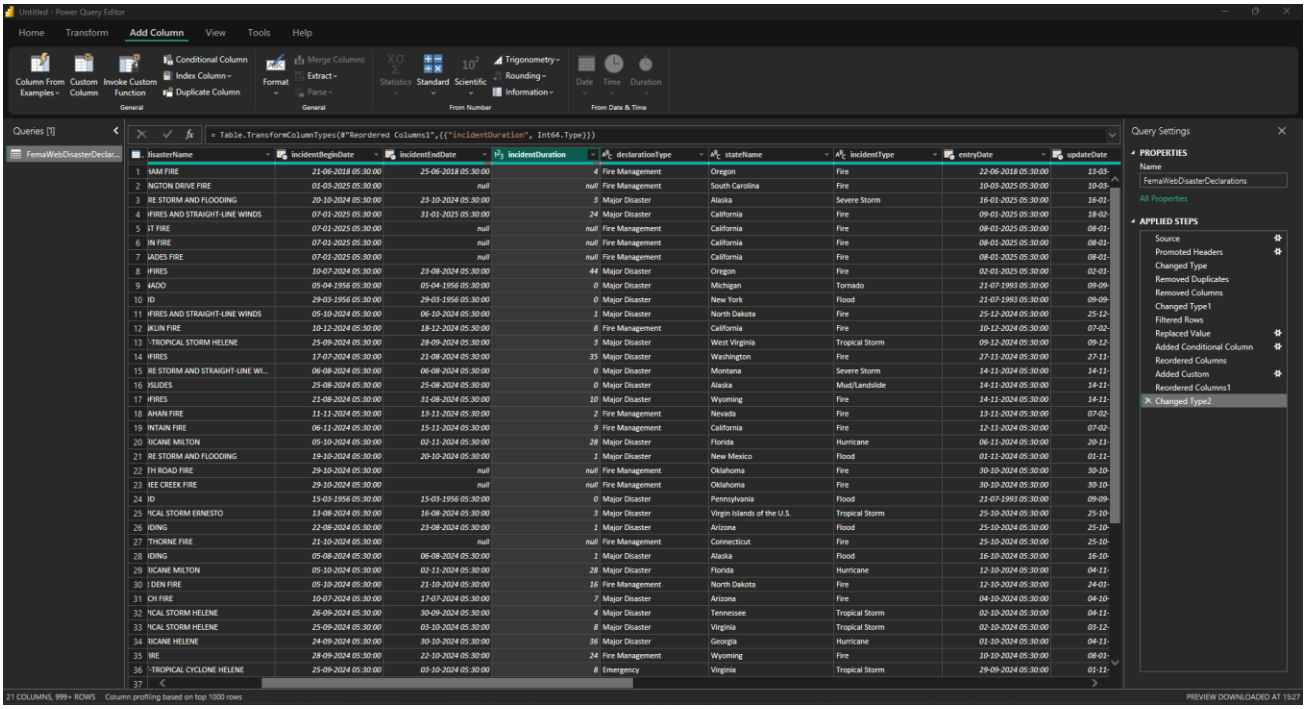


Fig 1.11 Added “incidentDuration” column

Change datatype of “incidentDuration” – whole number

3. Fiscal Year

In header, Add Column -> Custom Column

```
= if (Date.Month ([declarationDate]) >= 10)
  then (Date.Year([declarationDate]) + 1)
  else (Date.Year([declarationDate]) )
```

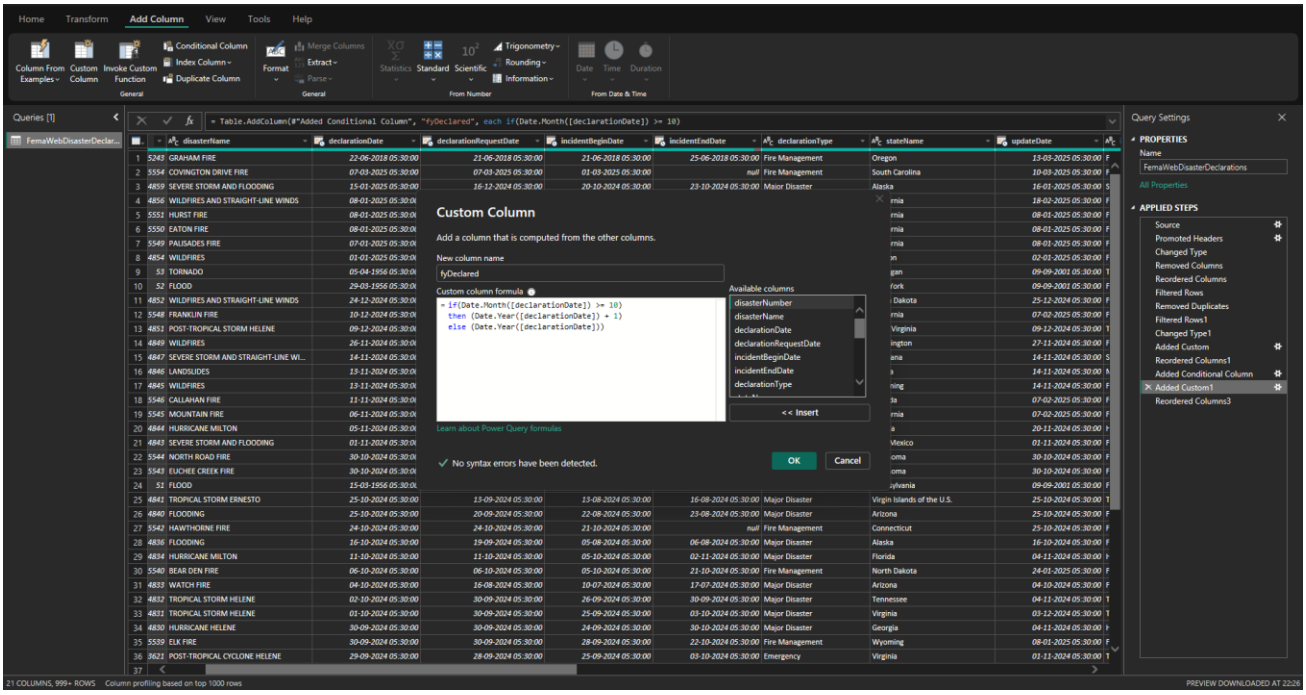


Fig 1.12 Custom Column pop-up for “fyDeclared” column

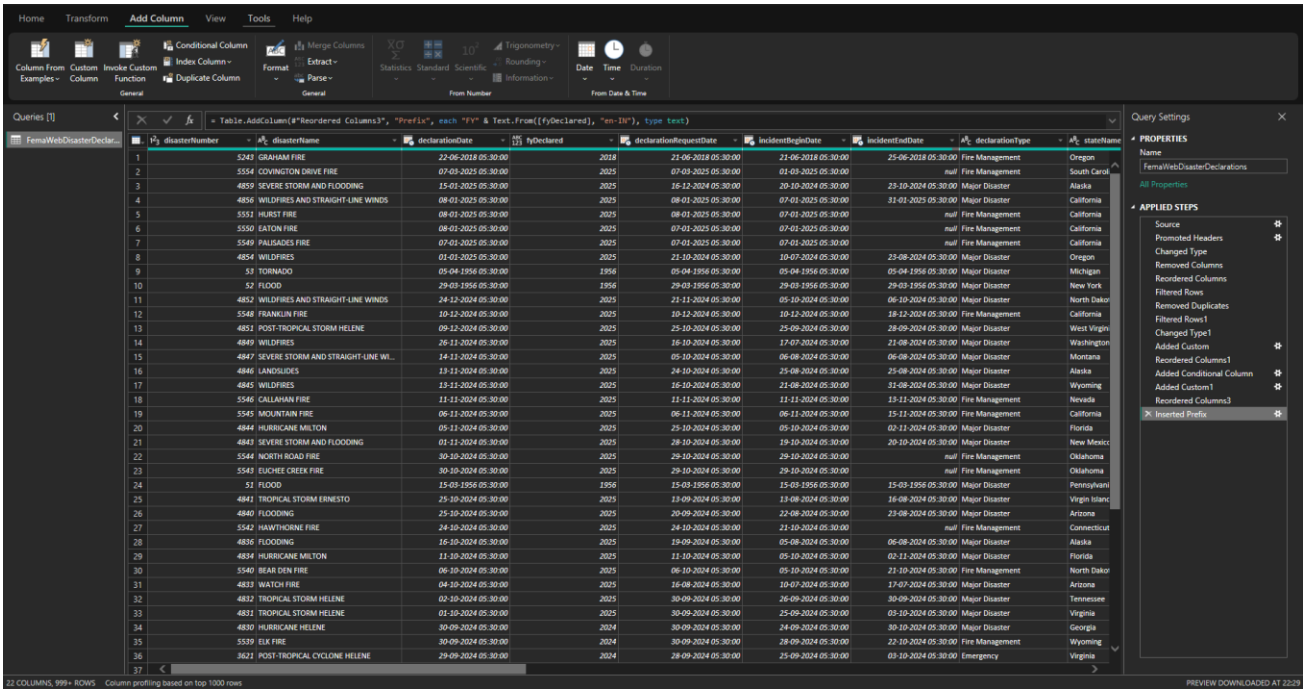
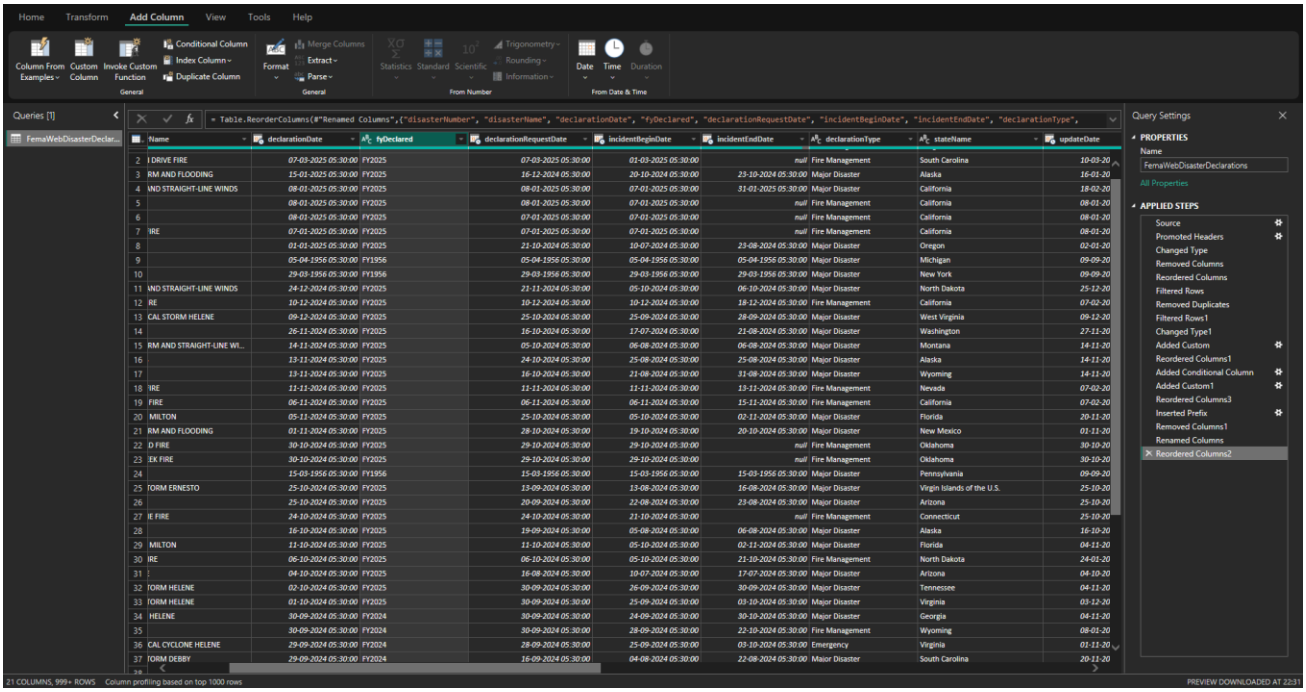


Fig 1.13 Added fyDeclared column

Select column -> Format -> Add Prefix -> FY

Delete previous column

Rename column



	Name	declarationDate	fyDeclared	declarationRequestDate	incidentBeginDate	incidentEndDate	declarationType	stateName	updateDate
2	I DRIVE FIRE	07-03-2025 05:30:00	FY2025	07-03-2025 05:30:00	01-03-2025 05:30:00	null	Fire Management	South Carolina	10-03-20
3	RM AND FLOODING	15-01-2025 05:30:00	FY2025	16-12-2024 05:30:00	20-10-2024 05:30:00	23-10-2024 05:30:00	Major Disaster	Alaska	16-01-20
4	ND STRAIGHT-LINE WINDS	08-01-2025 05:30:00	FY2025	08-01-2025 05:30:00	07-01-2025 05:30:00	31-01-2025 05:30:00	Major Disaster	California	18-02-20
5		08-01-2025 05:30:00	FY2025	08-01-2025 05:30:00	07-01-2025 05:30:00	null	Fire Management	California	08-01-20
6		08-01-2025 05:30:00	FY2025	07-01-2025 05:30:00	07-01-2025 05:30:00	null	Fire Management	California	08-01-20
7	IRE	07-01-2025 05:30:00	FY2025	07-01-2025 05:30:00	07-01-2025 05:30:00	null	Fire Management	California	08-01-20
8		01-01-2025 05:30:00	FY2025	21-10-2024 05:30:00	10-07-2024 05:30:00	23-08-2024 05:30:00	Major Disaster	Oregon	02-01-20
9		05-04-1956 05:30:00	FY1956	05-04-1956 05:30:00	05-04-1956 05:30:00	05-04-1956 05:30:00	Major Disaster	Michigan	09-09-20
10		29-03-1956 05:30:00	FY1956	29-03-1956 05:30:00	29-03-1956 05:30:00	29-03-1956 05:30:00	Major Disaster	New York	09-09-20
11	ND STRAIGHT-LINE WINDS	24-12-2024 05:30:00	FY2025	23-11-2024 05:30:00	05-10-2024 05:30:00	06-10-2024 05:30:00	Major Disaster	North Dakota	25-12-20
12	RE	10-12-2024 05:30:00	FY2025	10-12-2024 05:30:00	10-12-2024 05:30:00	18-12-2024 05:30:00	Fire Management	California	07-02-20
13	CAL STORM HELENE	09-12-2024 05:30:00	FY2025	25-10-2024 05:30:00	25-09-2024 05:30:00	28-09-2024 05:30:00	Major Disaster	West Virginia	09-12-20
14		26-11-2024 05:30:00	FY2025	16-10-2024 05:30:00	17-07-2024 05:30:00	21-08-2024 05:30:00	Major Disaster	Washington	27-11-20
15	RM AND STRAIGHT-LINE WI...	14-11-2024 05:30:00	FY2025	05-10-2024 05:30:00	06-08-2024 05:30:00	06-08-2024 05:30:00	Major Disaster	Montana	14-11-20
16		13-11-2024 05:30:00	FY2025	24-10-2024 05:30:00	25-08-2024 05:30:00	25-08-2024 05:30:00	Major Disaster	Alaska	14-11-20
17		13-11-2024 05:30:00	FY2025	16-10-2024 05:30:00	21-08-2024 05:30:00	31-08-2024 05:30:00	Major Disaster	Wyoming	14-11-20
18	IRE	11-11-2024 05:30:00	FY2025	11-11-2024 05:30:00	11-11-2024 05:30:00	13-11-2024 05:30:00	Fire Management	Nevada	07-02-20
19	FIRE	06-11-2024 05:30:00	FY2025	06-11-2024 05:30:00	06-11-2024 05:30:00	15-11-2024 05:30:00	Fire Management	California	07-02-20
20	MILTON	05-11-2024 05:30:00	FY2025	25-10-2024 05:30:00	05-10-2024 05:30:00	02-11-2024 05:30:00	Major Disaster	Florida	20-11-20
21	RM AND FLOODING	01-11-2024 05:30:00	FY2025	28-10-2024 05:30:00	19-10-2024 05:30:00	20-10-2024 05:30:00	Major Disaster	New Mexico	01-11-20
22	D FIRE	30-10-2024 05:30:00	FY2025	29-10-2024 05:30:00	29-10-2024 05:30:00	null	Fire Management	Oklahoma	30-10-20
23	EK FIRE	30-10-2024 05:30:00	FY2025	29-10-2024 05:30:00	29-10-2024 05:30:00	null	Fire Management	Oklahoma	30-10-20
24		15-03-1956 05:30:00	FY1956	15-03-1956 05:30:00	15-03-1956 05:30:00	15-03-1956 05:30:00	Major Disaster	Pennsylvania	09-09-20
25	TORM ERNESTO	25-10-2024 05:30:00	FY2025	13-09-2024 05:30:00	13-08-2024 05:30:00	16-08-2024 05:30:00	Major Disaster	Virgin Islands of the U.S.	25-10-20
26		25-10-2024 05:30:00	FY2025	20-09-2024 05:30:00	22-08-2024 05:30:00	23-08-2024 05:30:00	Major Disaster	Arizona	25-10-20
27	E FIRE	24-10-2024 05:30:00	FY2025	24-10-2024 05:30:00	21-10-2024 05:30:00	null	Fire Management	Connecticut	25-10-20
28		16-10-2024 05:30:00	FY2025	19-09-2024 05:30:00	05-08-2024 05:30:00	06-08-2024 05:30:00	Major Disaster	Alaska	16-10-20
29	MILTON	11-10-2024 05:30:00	FY2025	11-10-2024 05:30:00	05-10-2024 05:30:00	02-11-2024 05:30:00	Major Disaster	Florida	04-11-20
30	IRE	06-10-2024 05:30:00	FY2025	06-10-2024 05:30:00	05-10-2024 05:30:00	21-10-2024 05:30:00	Fire Management	North Dakota	24-01-20
31		04-10-2024 05:30:00	FY2025	16-08-2024 05:30:00	10-07-2024 05:30:00	17-07-2024 05:30:00	Major Disaster	Arizona	04-10-20
32	TORM HELENE	02-10-2024 05:30:00	FY2025	30-09-2024 05:30:00	26-09-2024 05:30:00	30-09-2024 05:30:00	Major Disaster	Tennessee	04-11-20
33	TORM HELENE	01-10-2024 05:30:00	FY2025	30-09-2024 05:30:00	25-09-2024 05:30:00	03-10-2024 05:30:00	Major Disaster	Virginia	03-12-20
34	HELENE	30-09-2024 05:30:00	FY2024	30-09-2024 05:30:00	24-09-2024 05:30:00	30-10-2024 05:30:00	Major Disaster	Georgia	04-11-20
35		30-09-2024 05:30:00	FY2024	30-09-2024 05:30:00	28-09-2024 05:30:00	22-10-2024 05:30:00	Fire Management	Wyoming	08-01-20
36	CAL CYCLONE HELENE	29-09-2024 05:30:00	FY2024	28-09-2024 05:30:00	25-09-2024 05:30:00	03-10-2024 05:30:00	Emergency	Virginia	01-11-20
37	TORM DEBBY	29-09-2024 05:30:00	FY2024	16-09-2024 05:30:00	04-08-2024 05:30:00	22-08-2024 05:30:00	Major Disaster	South Carolina	20-11-20

Fig 1.14 Final fyDeclared column